

Modélisation cinématique d'un système de commande de soupape

Sources : - J LEGOFF / P.PROVOT

Retravaillé par D GONCALVES

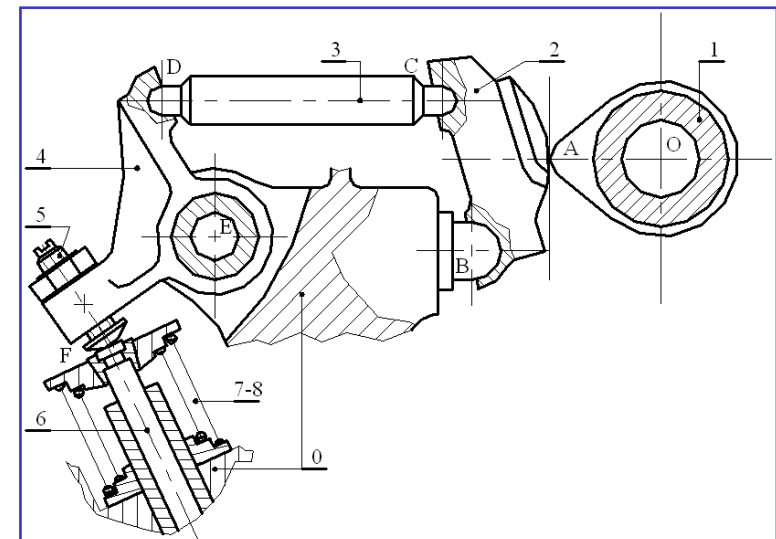
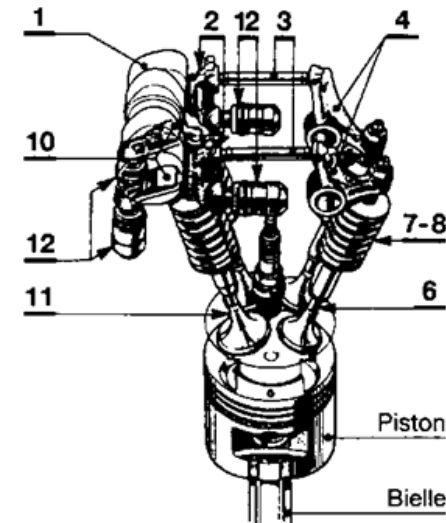
1. Présentation

Le dessin en perspective représente le mécanisme de commande des quatre soupapes d'un cylindre de moteur à explosion.

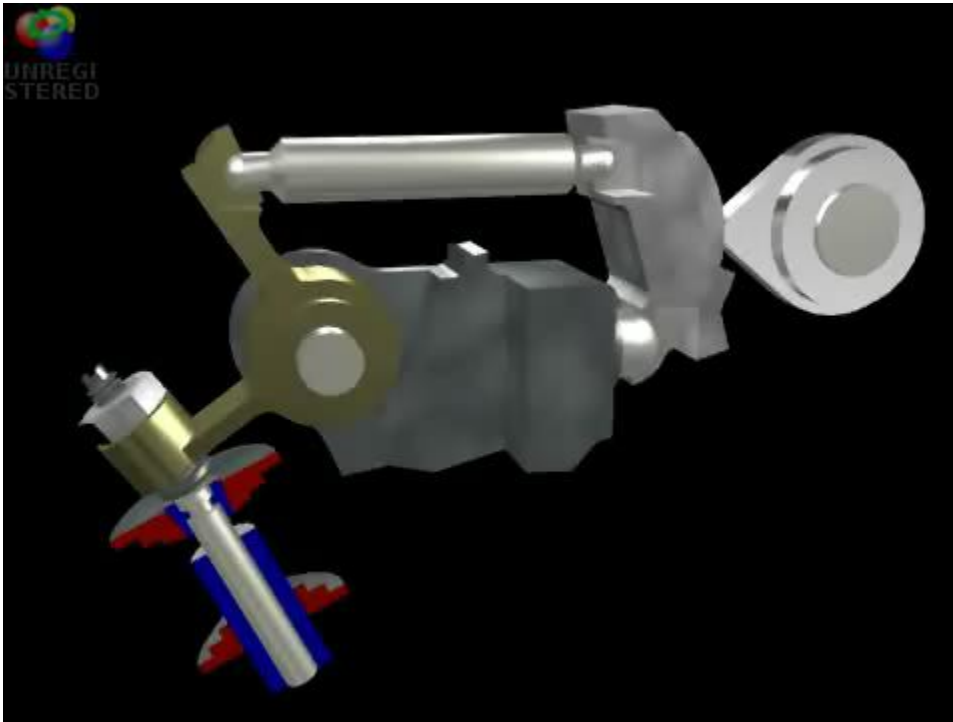
Pour ouvrir l'une des soupapes, la came 1 agit en A sur le linguet 2 qui transmet une action mécanique à la soupape 6 par l'intermédiaire de la tige 3, du culbuteur 4, du poussoir réglable 5 en comprimant le ressort 7-8.

L'examen de ce mécanisme montre que toutes ces pièces et les efforts appliqués ont un plan de symétrie en commun.

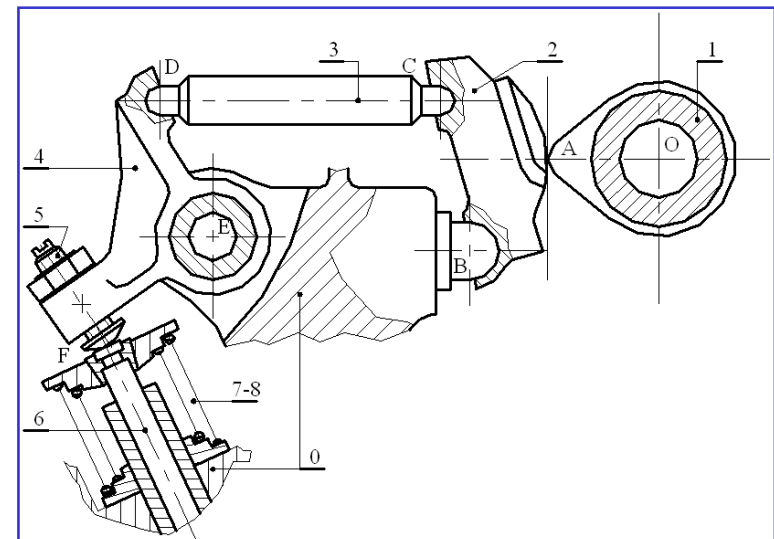
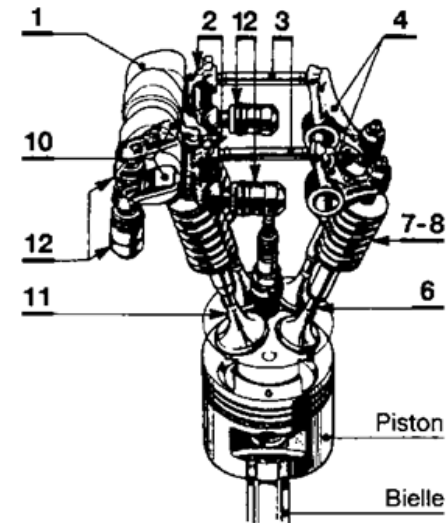
Mécanisme de commande de soupapes



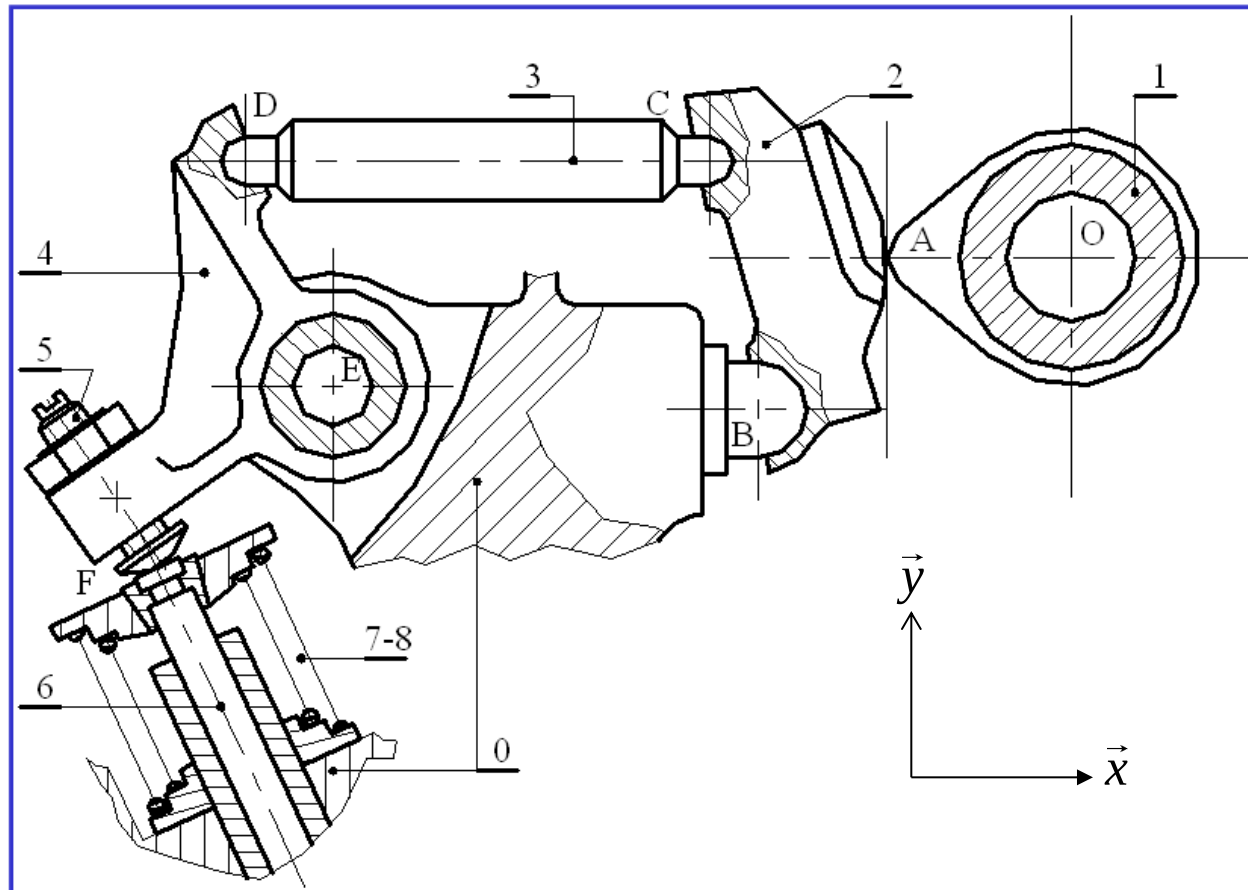
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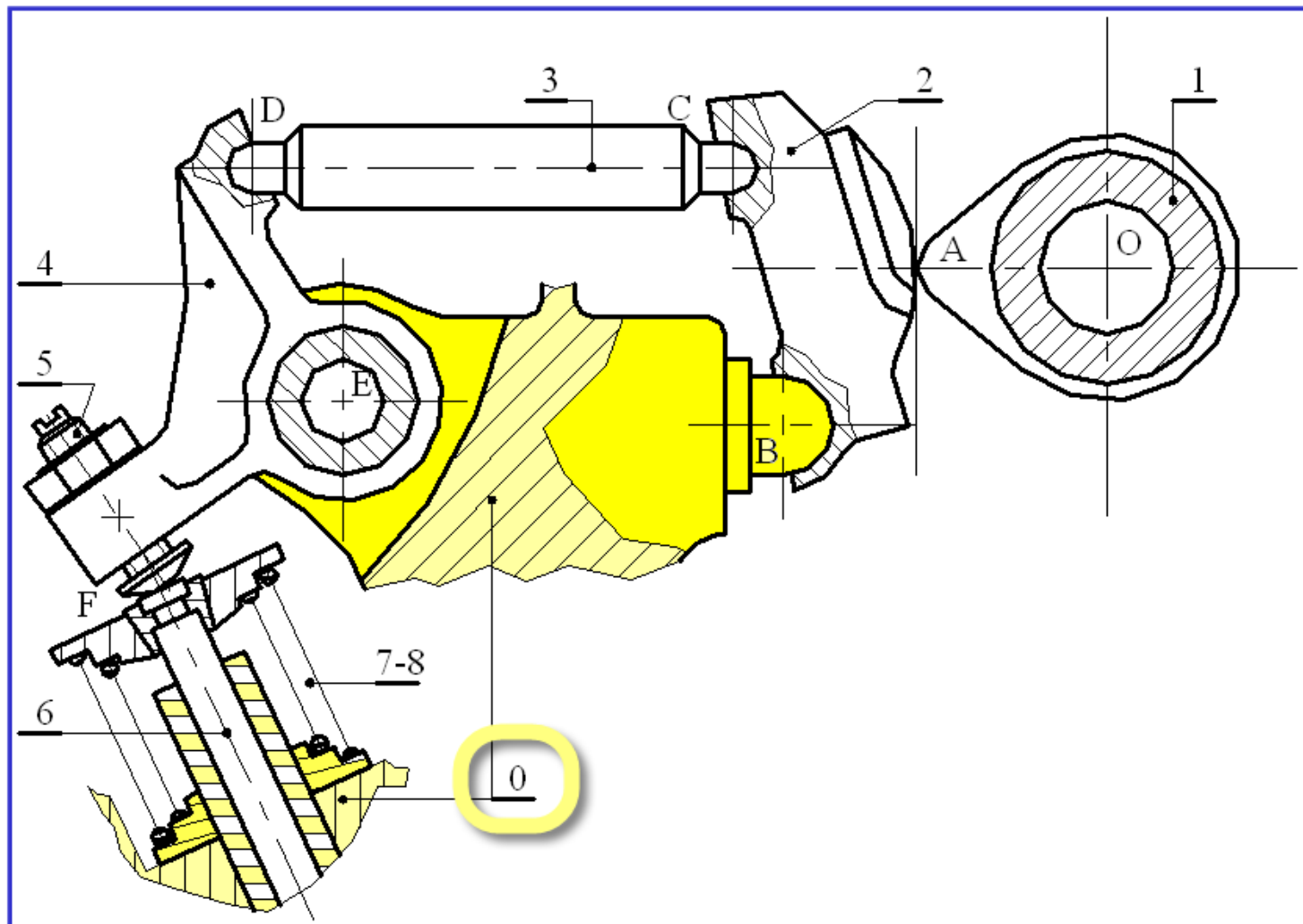
Mécanisme de commande de soupapes



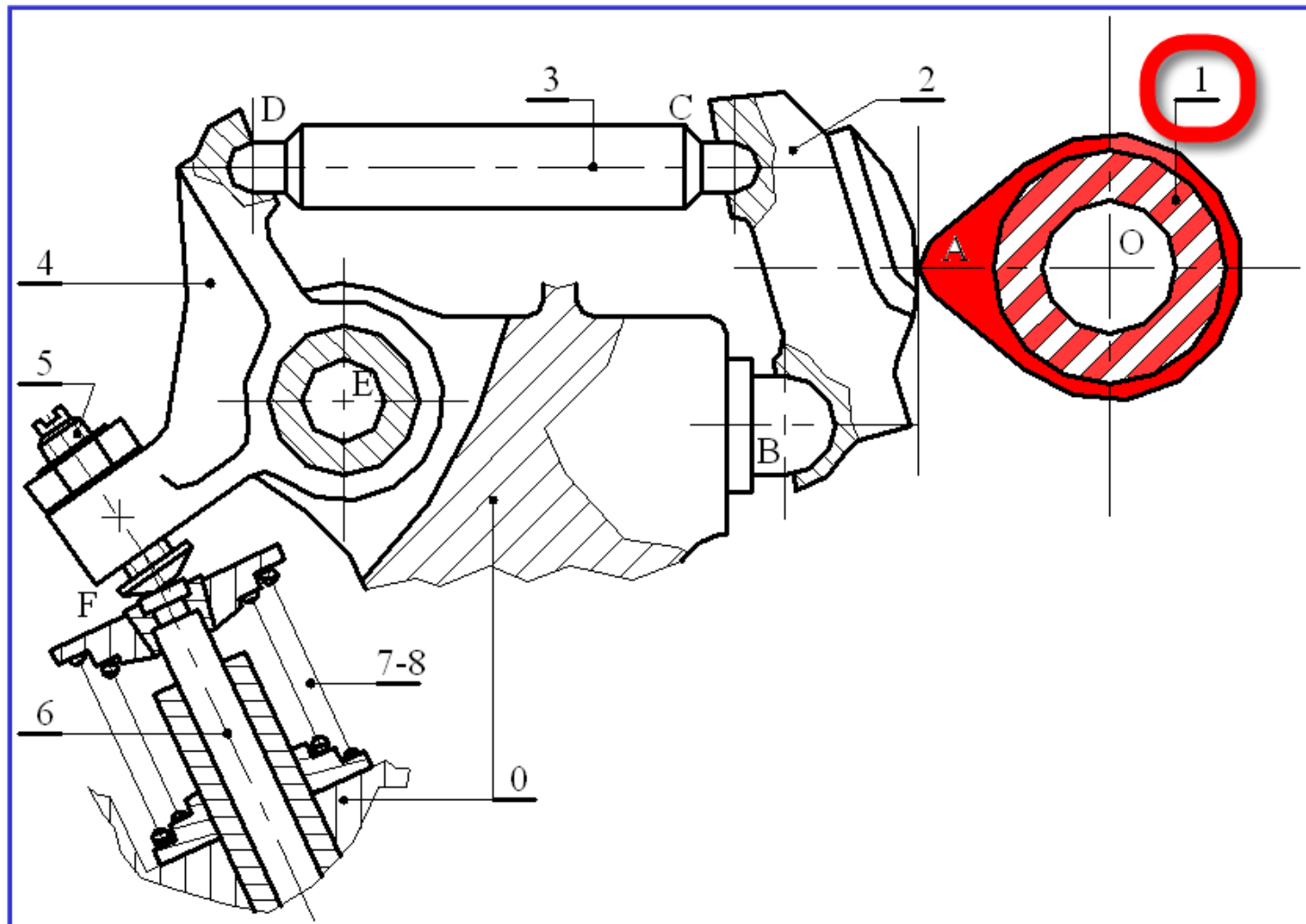
Etablir le graphe des liaisons et dessiner le schéma cinématique du système de commande de soupape (dans le plan $(\vec{x}, \vec{y})$)



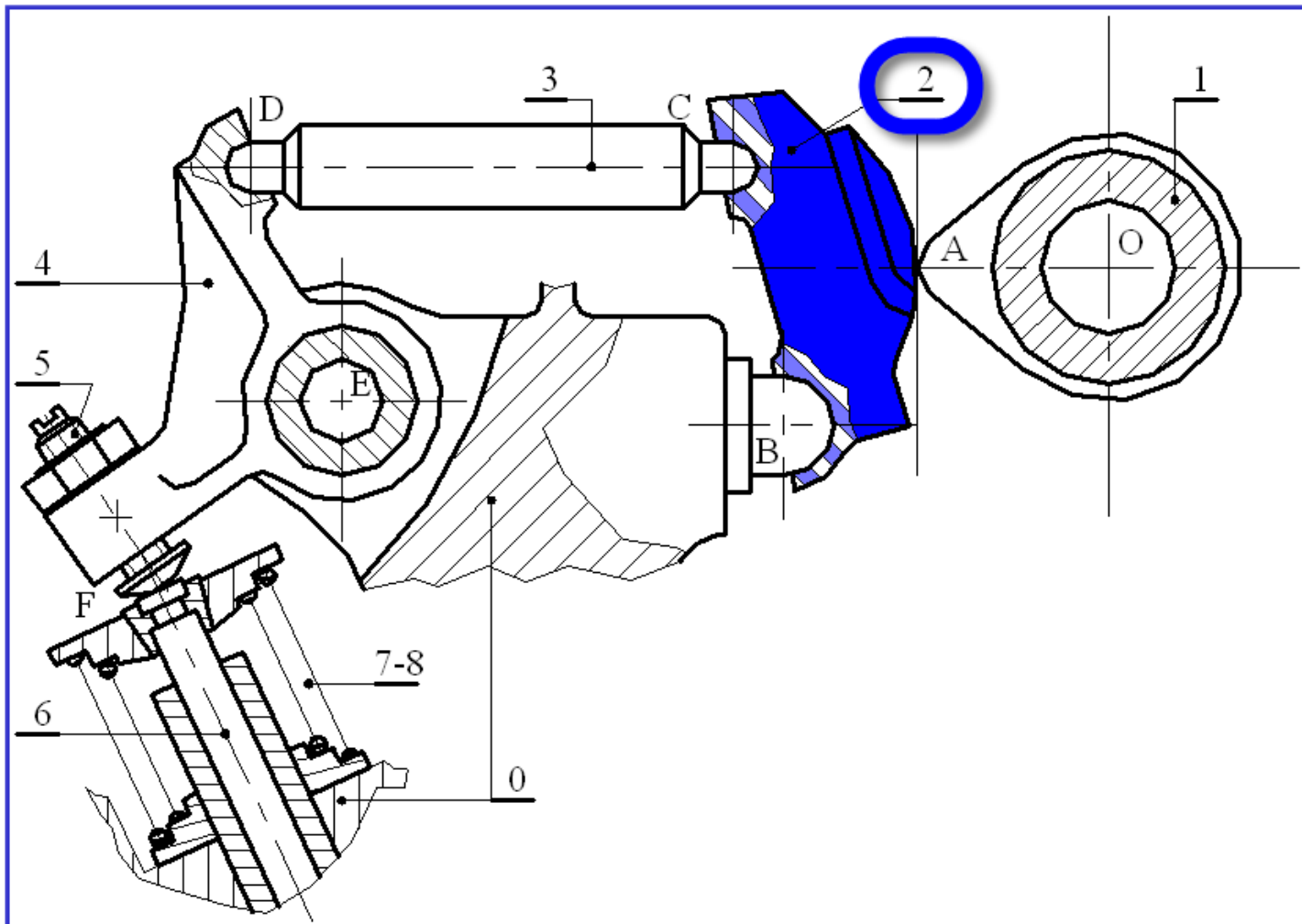
Recherche des classes d'équivalence (ou solides cinématiques)



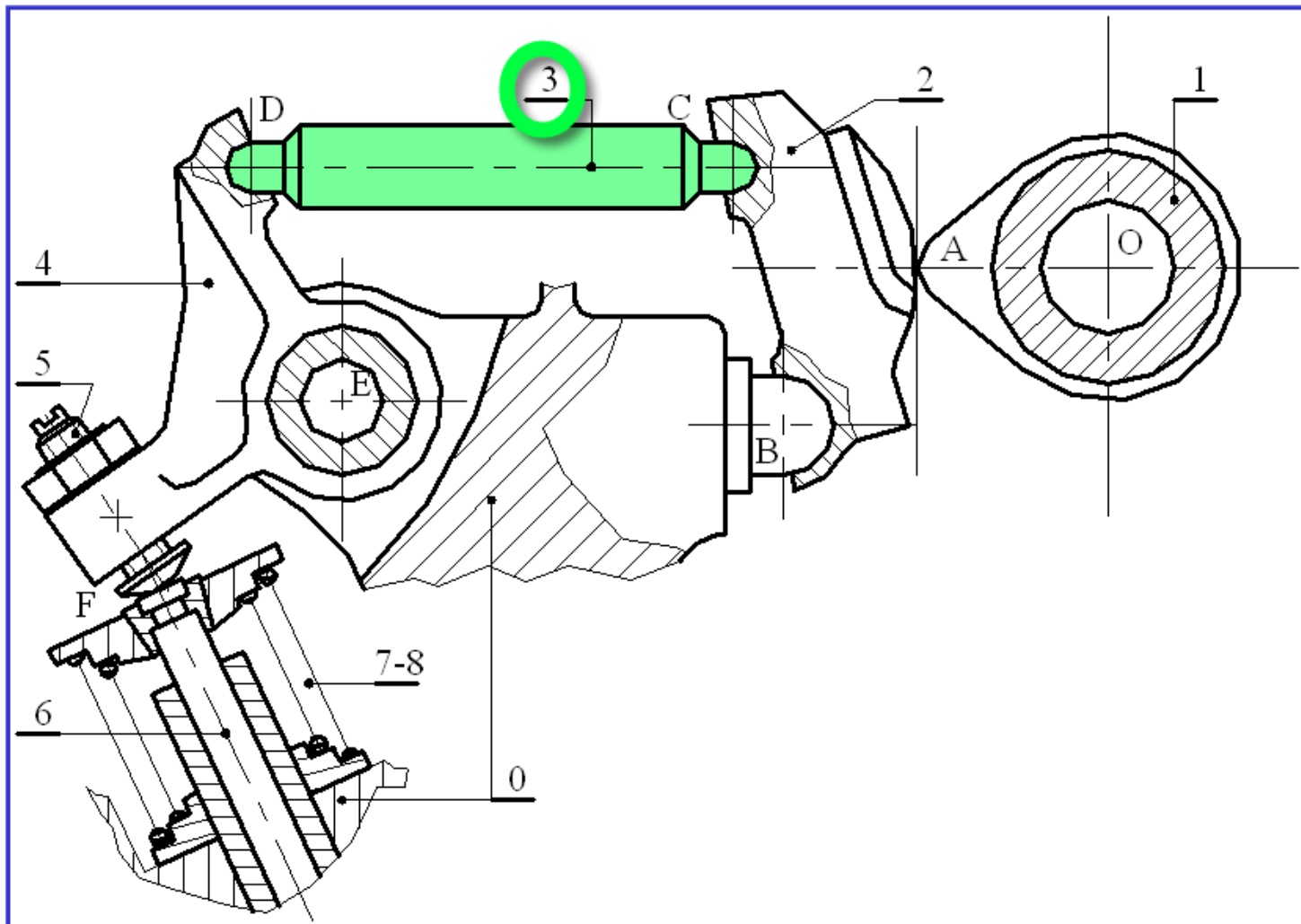
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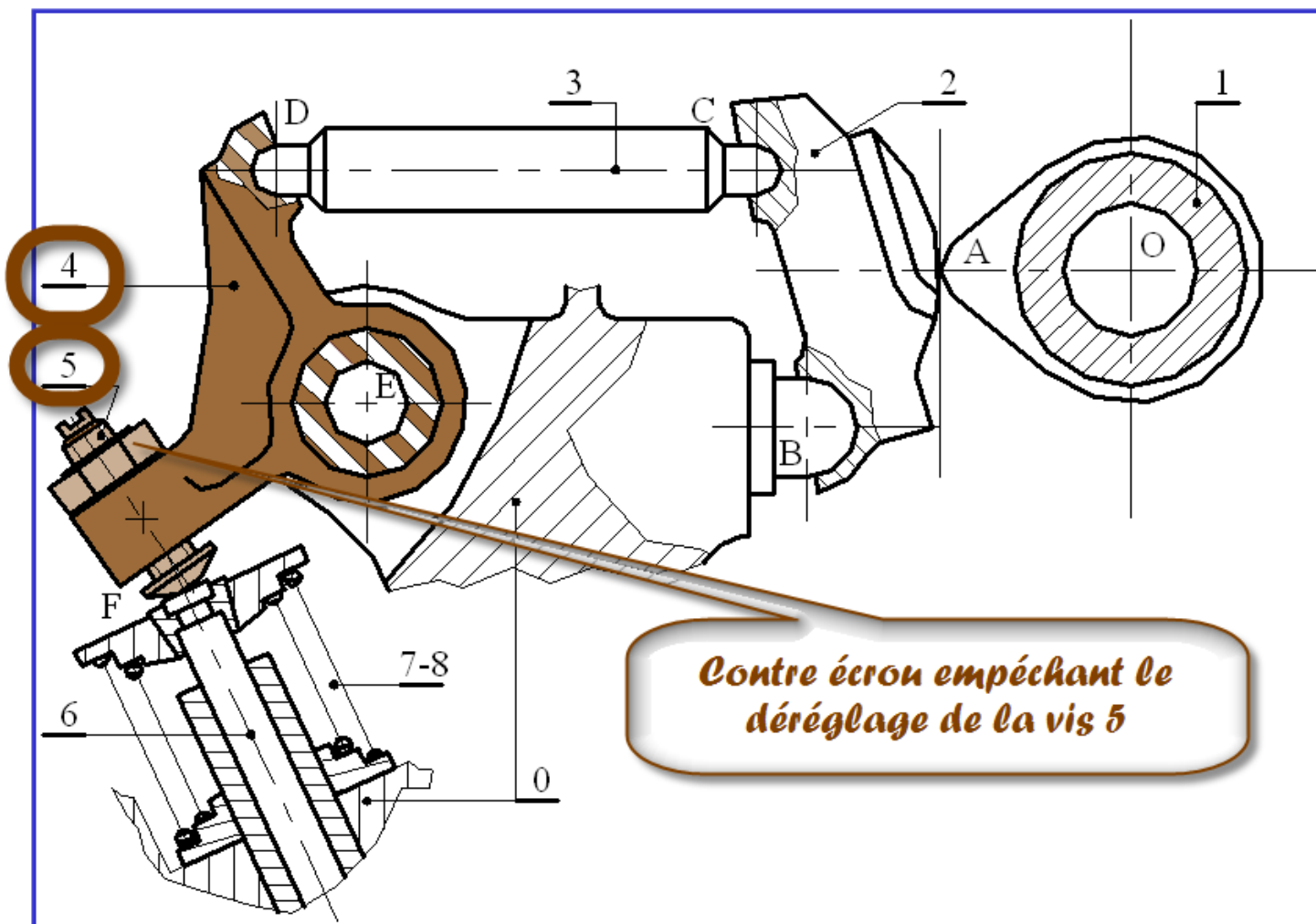
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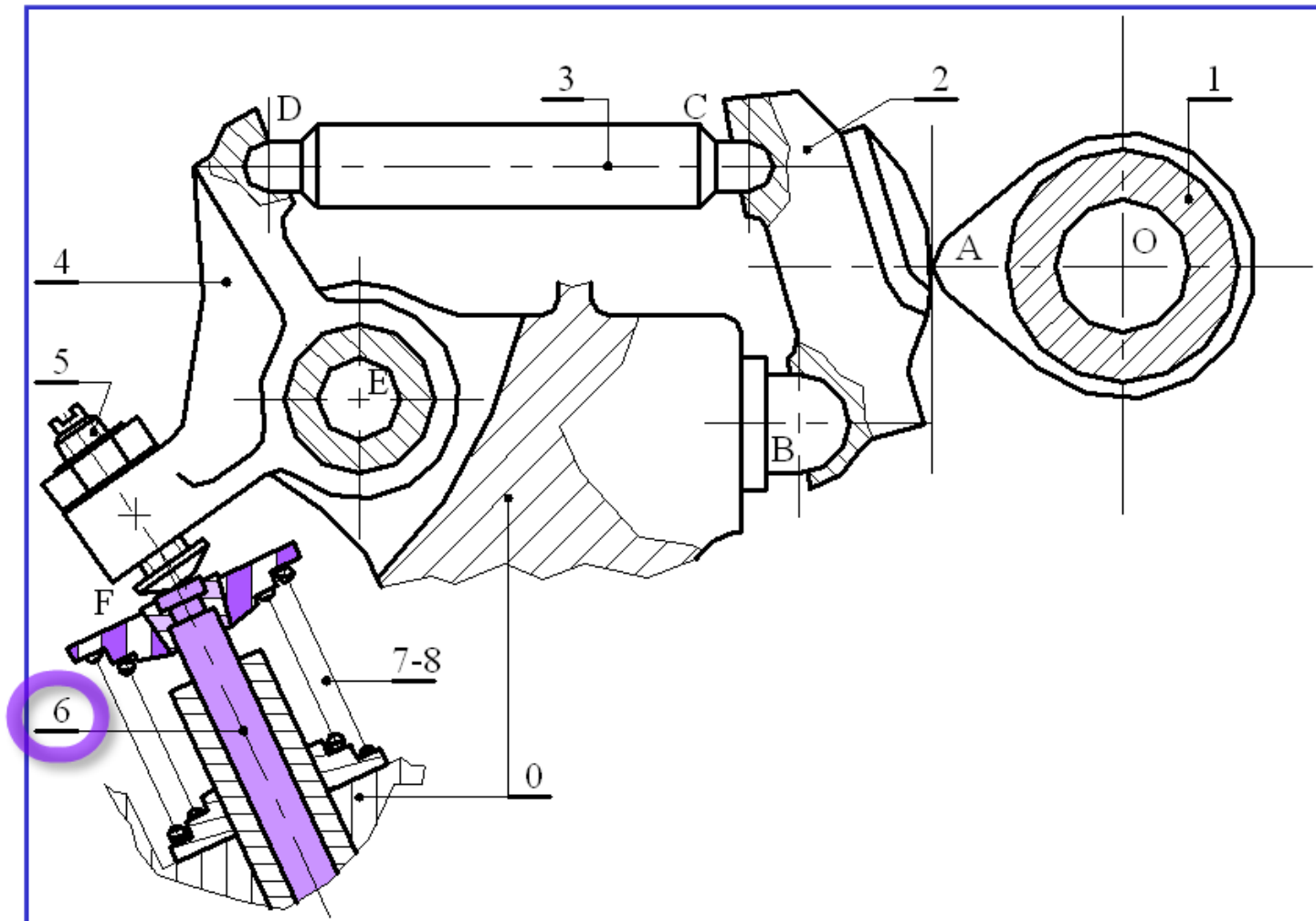
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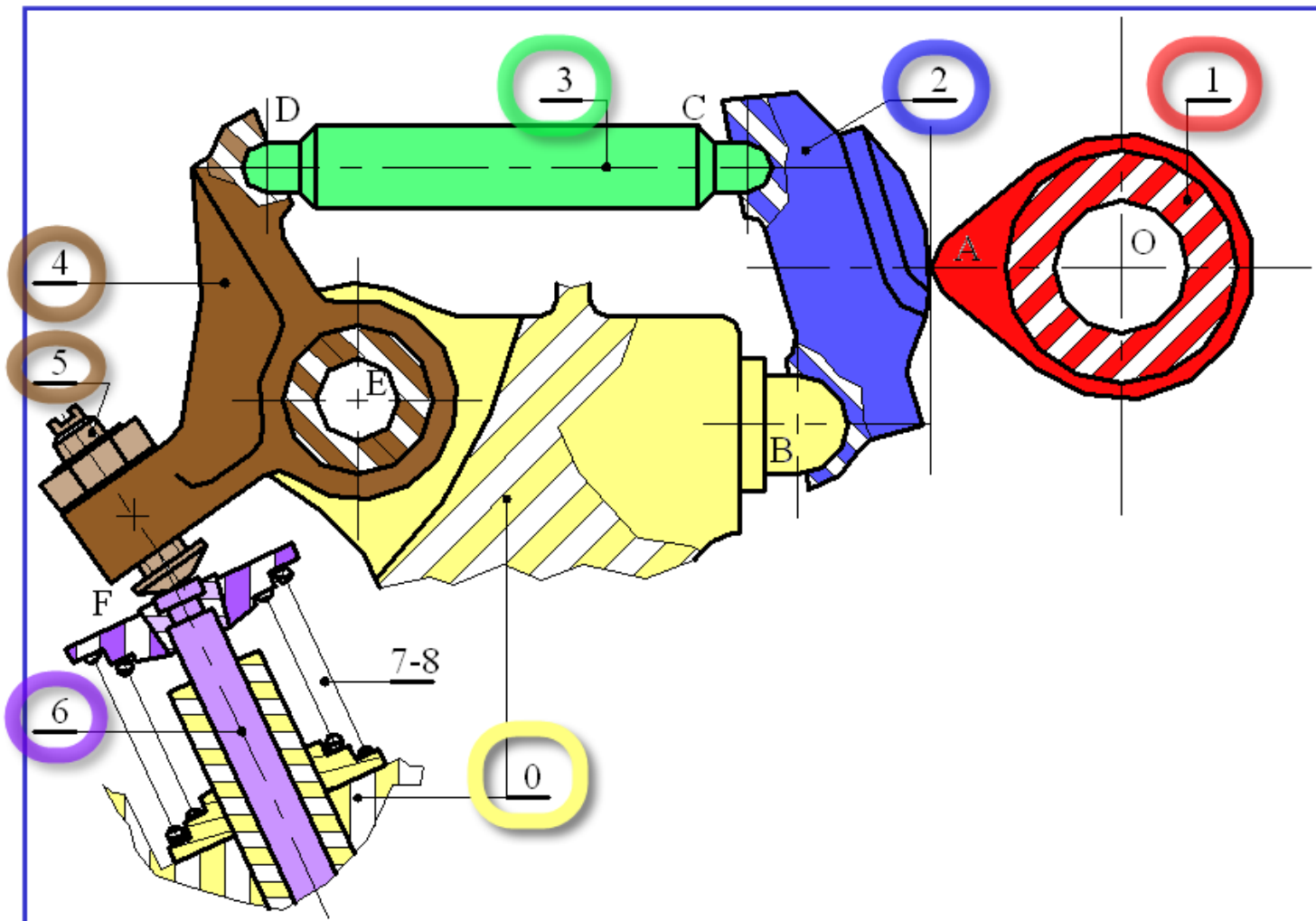
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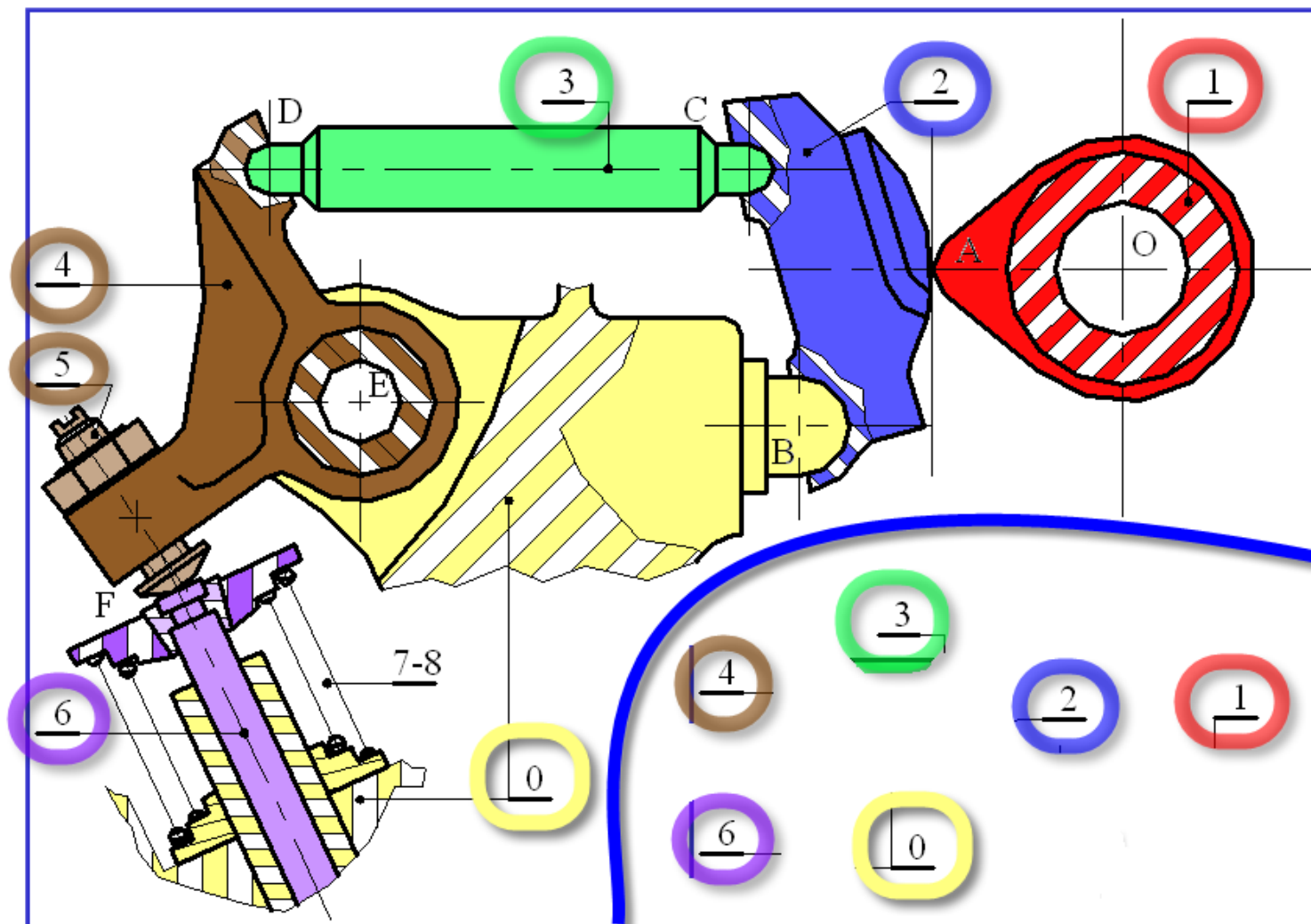
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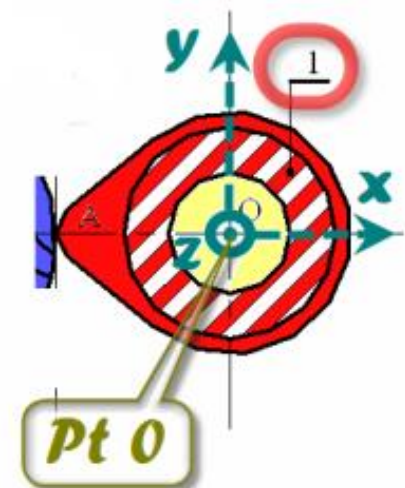
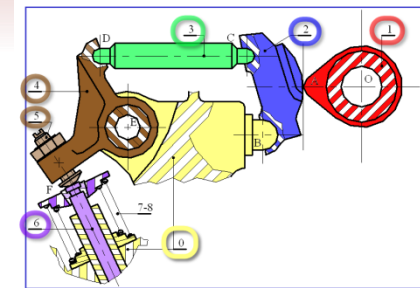
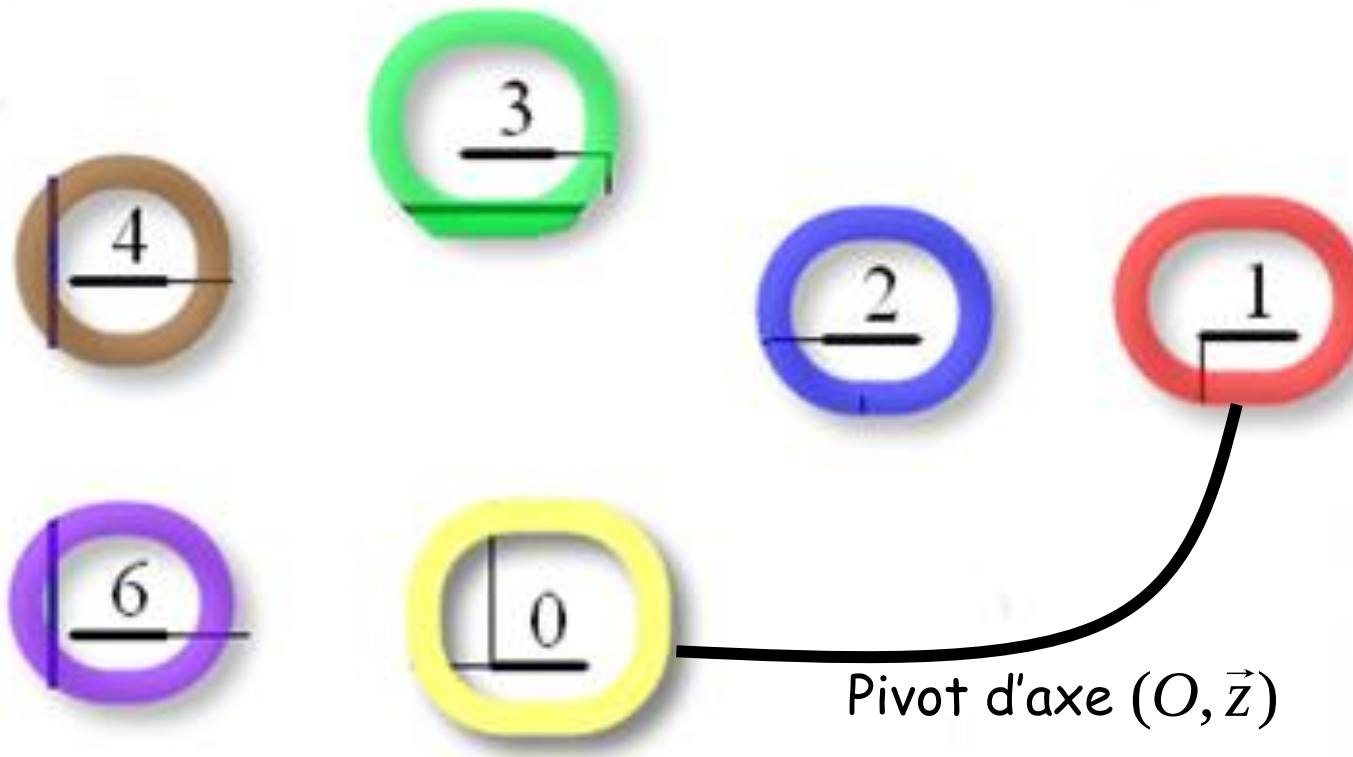
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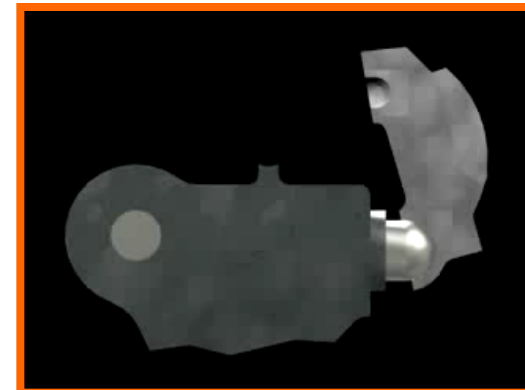
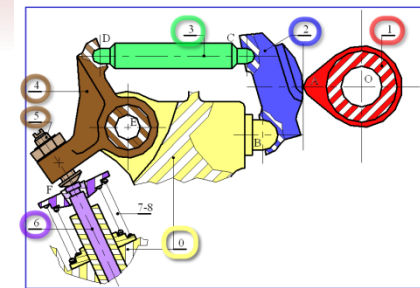
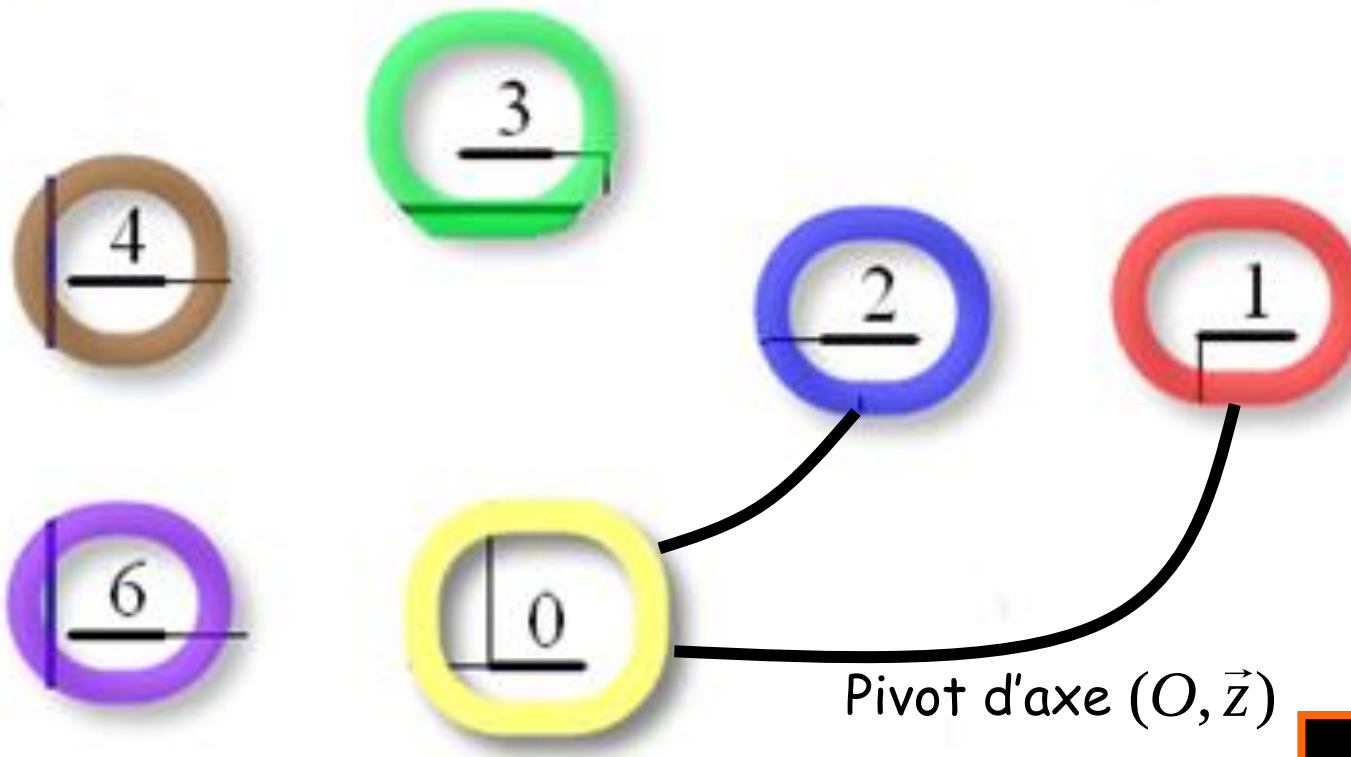
Graphe des liaisons



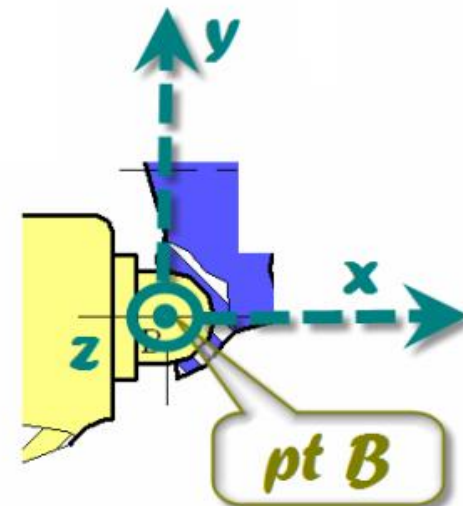
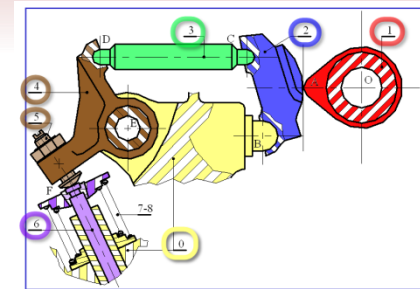
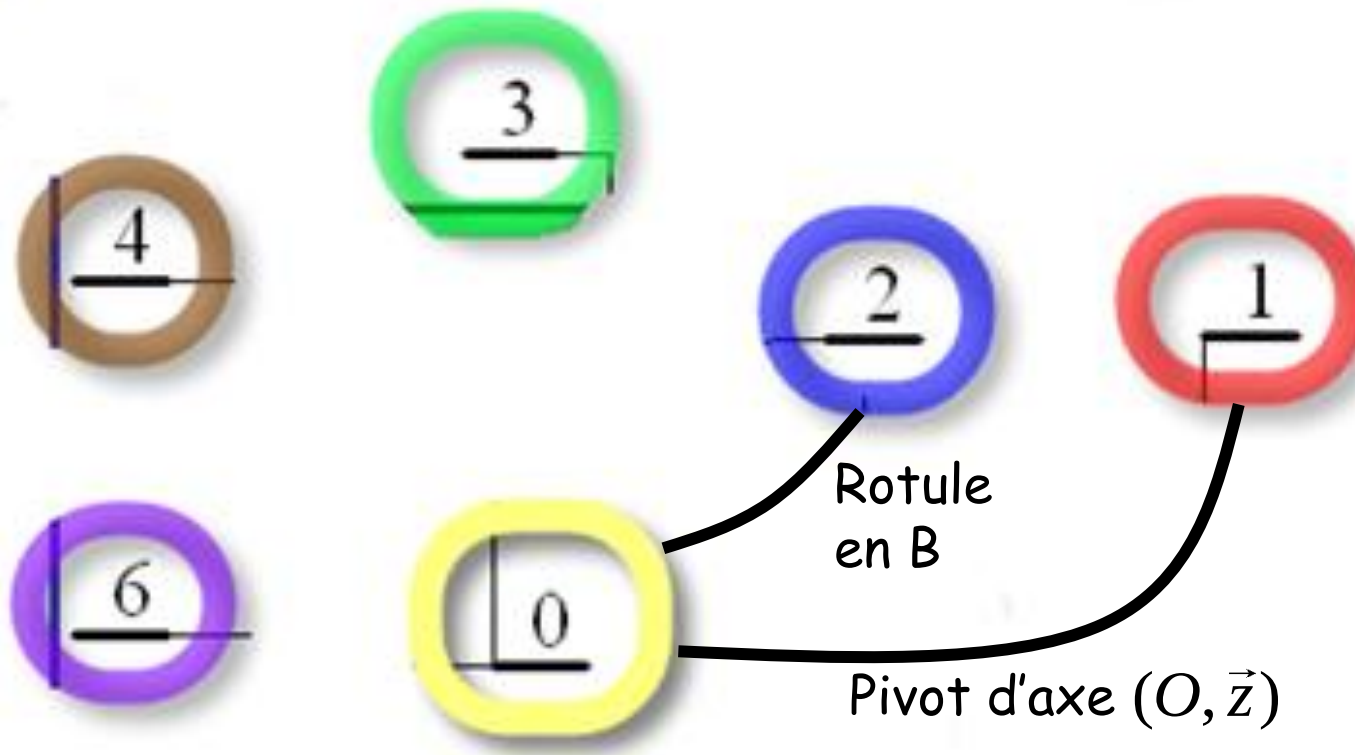
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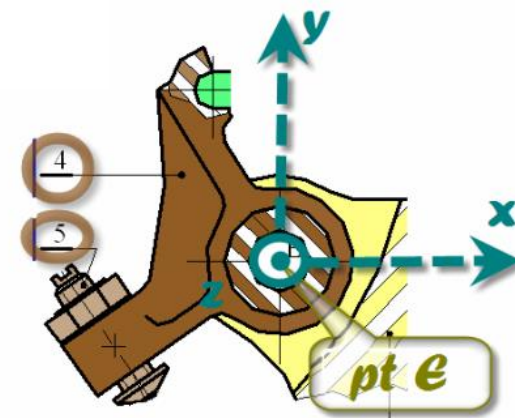
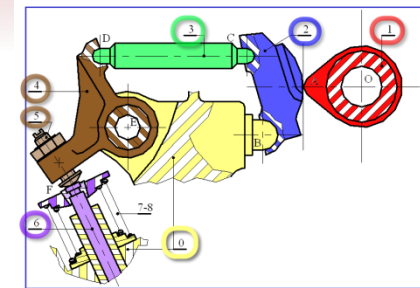
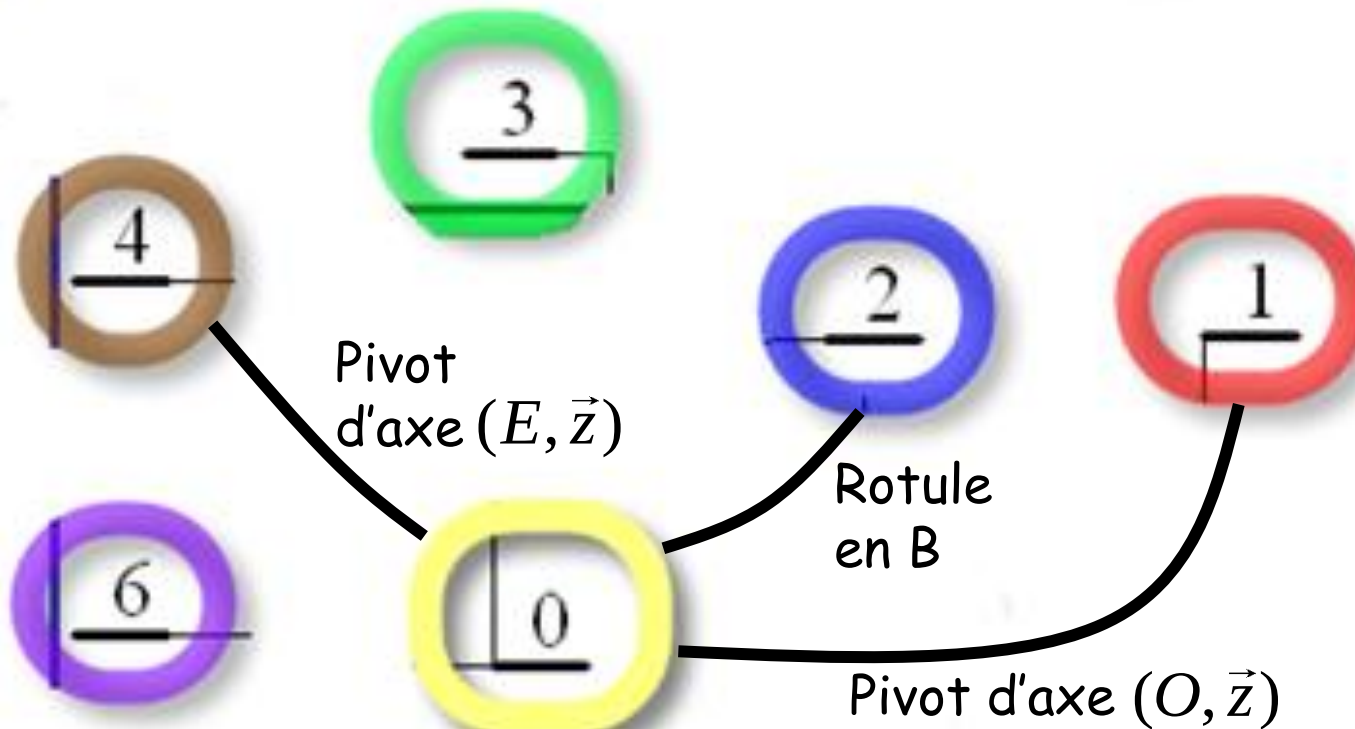
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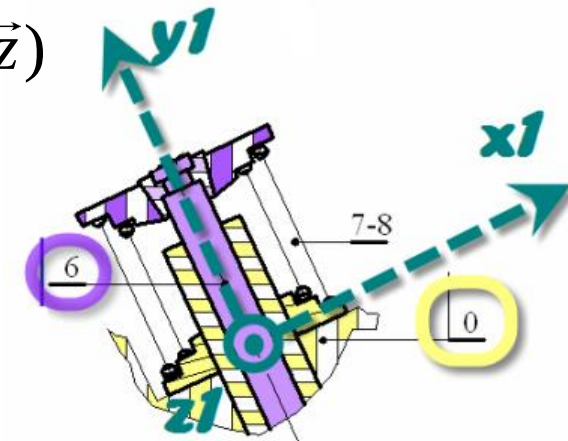
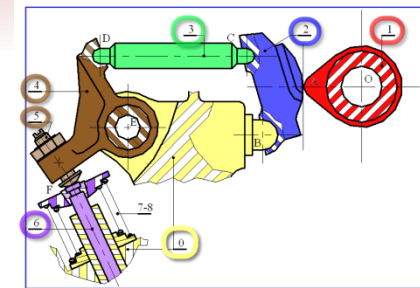
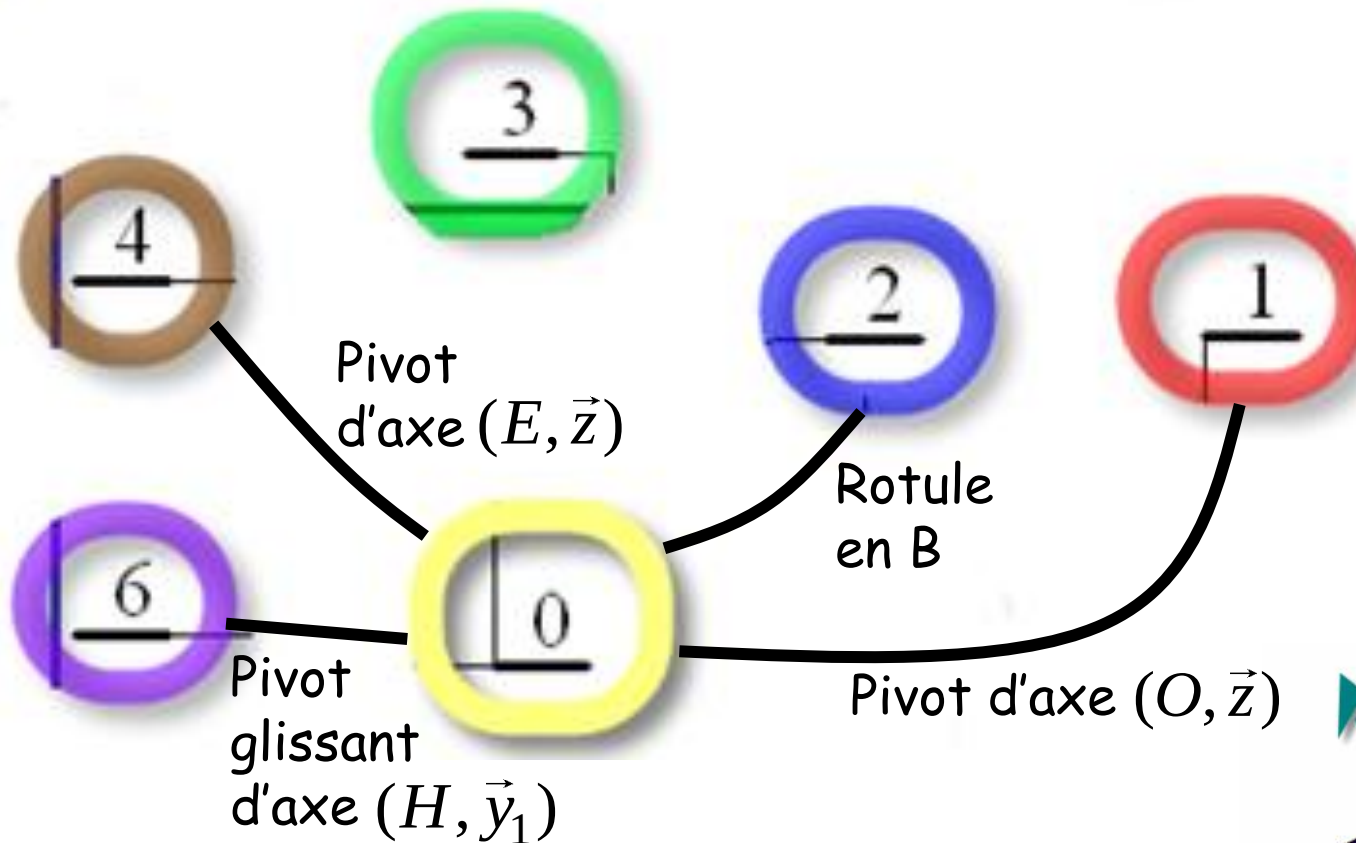
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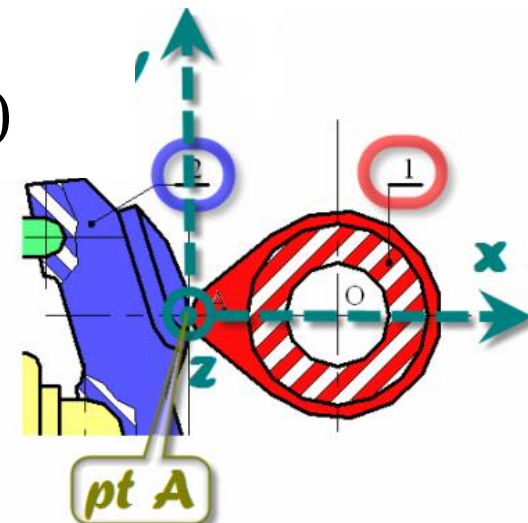
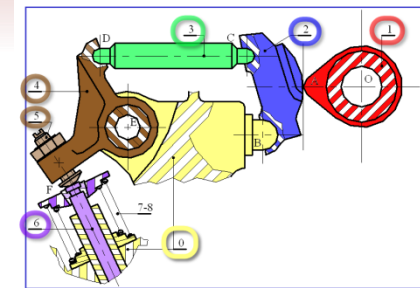
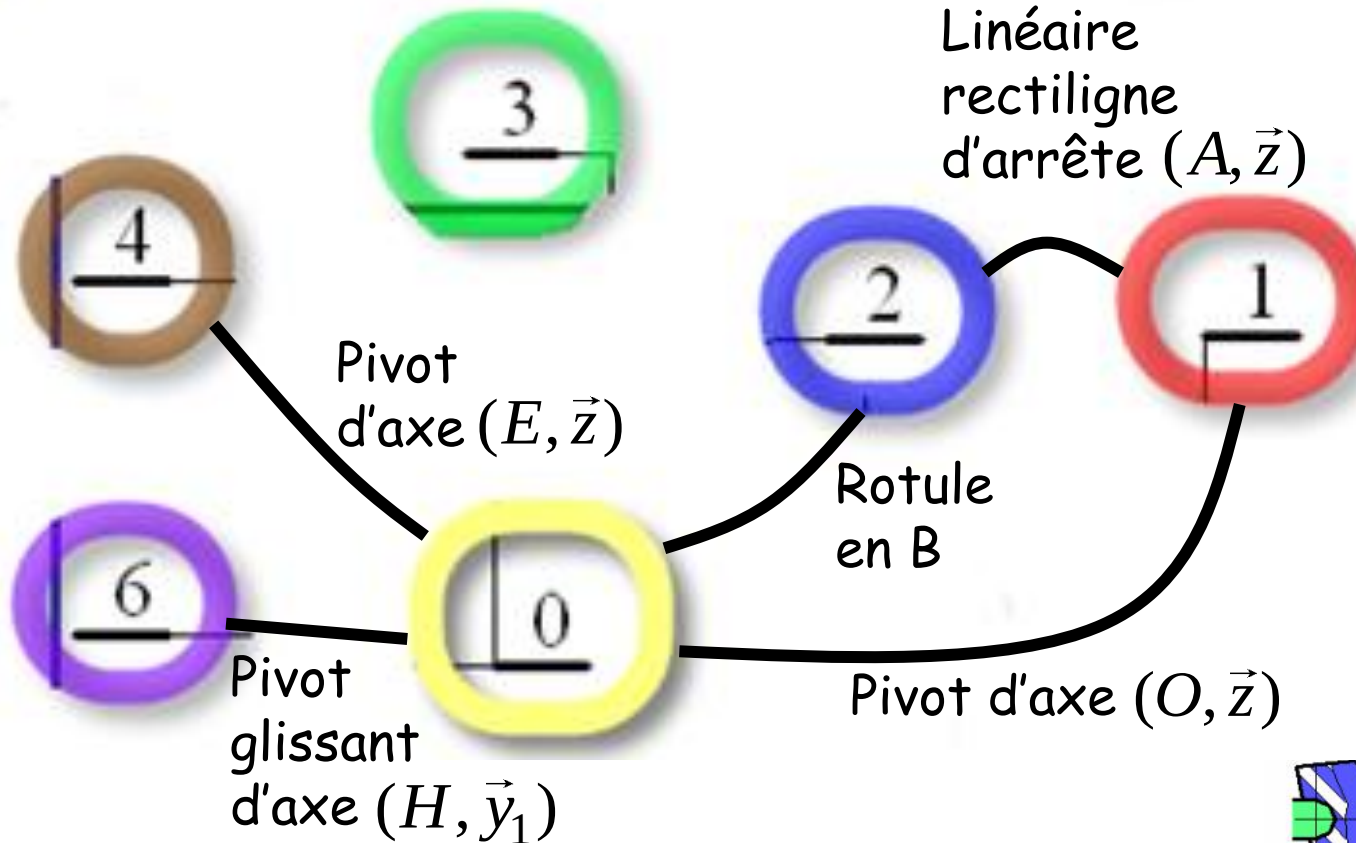
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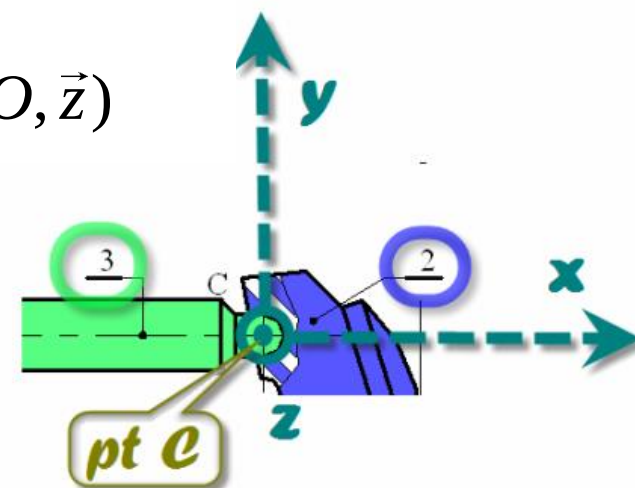
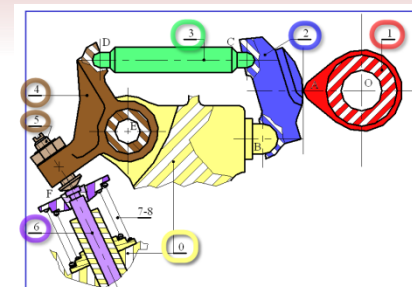
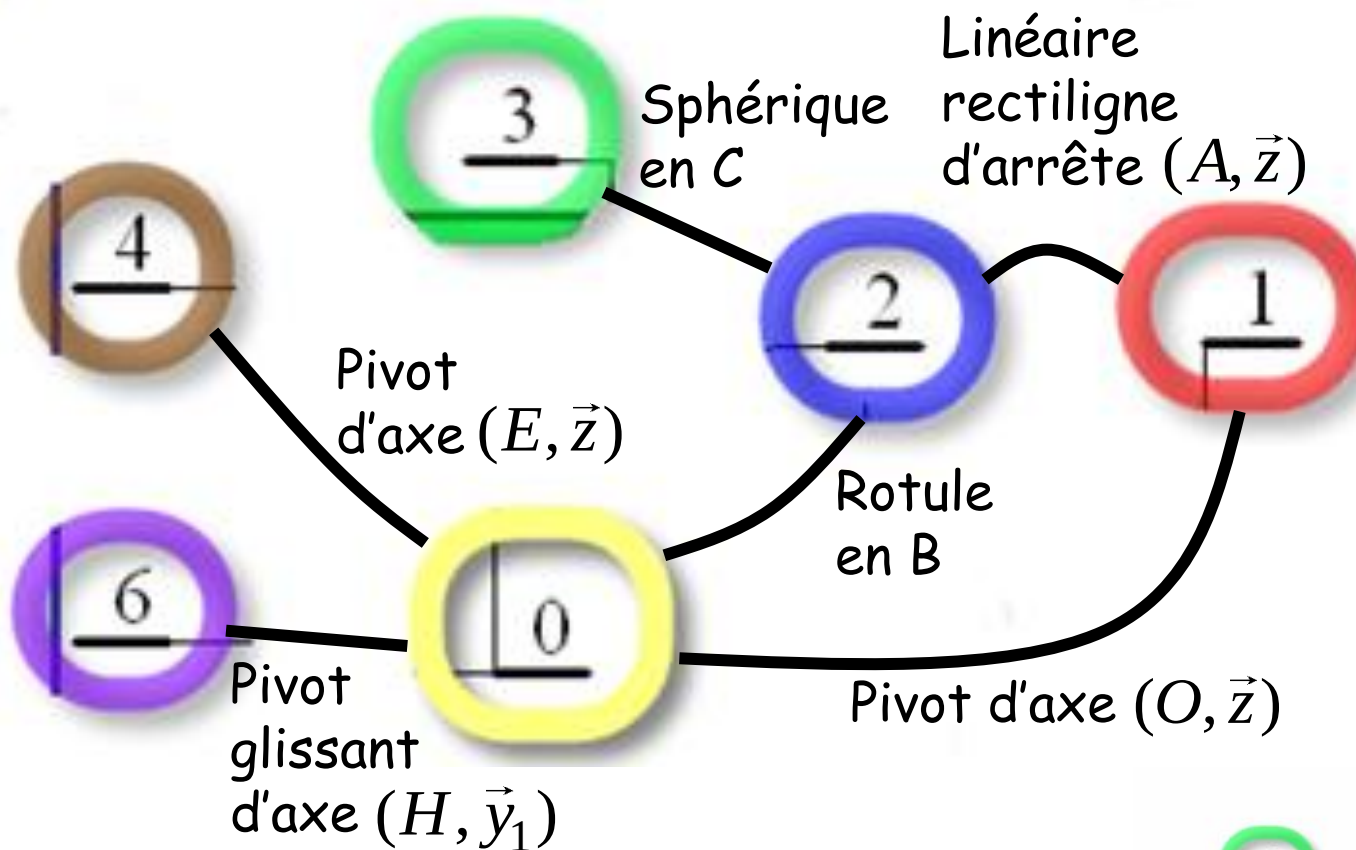
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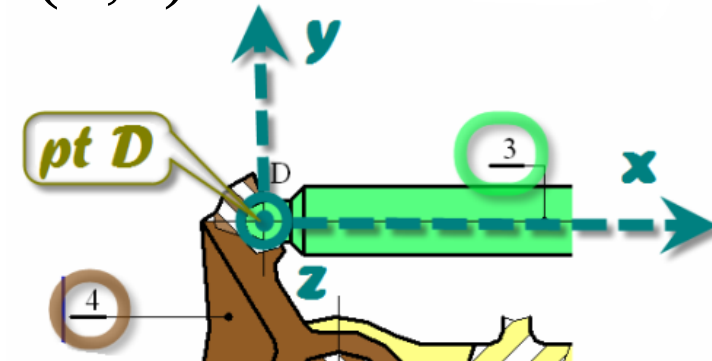
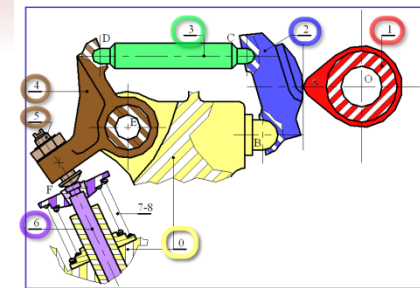
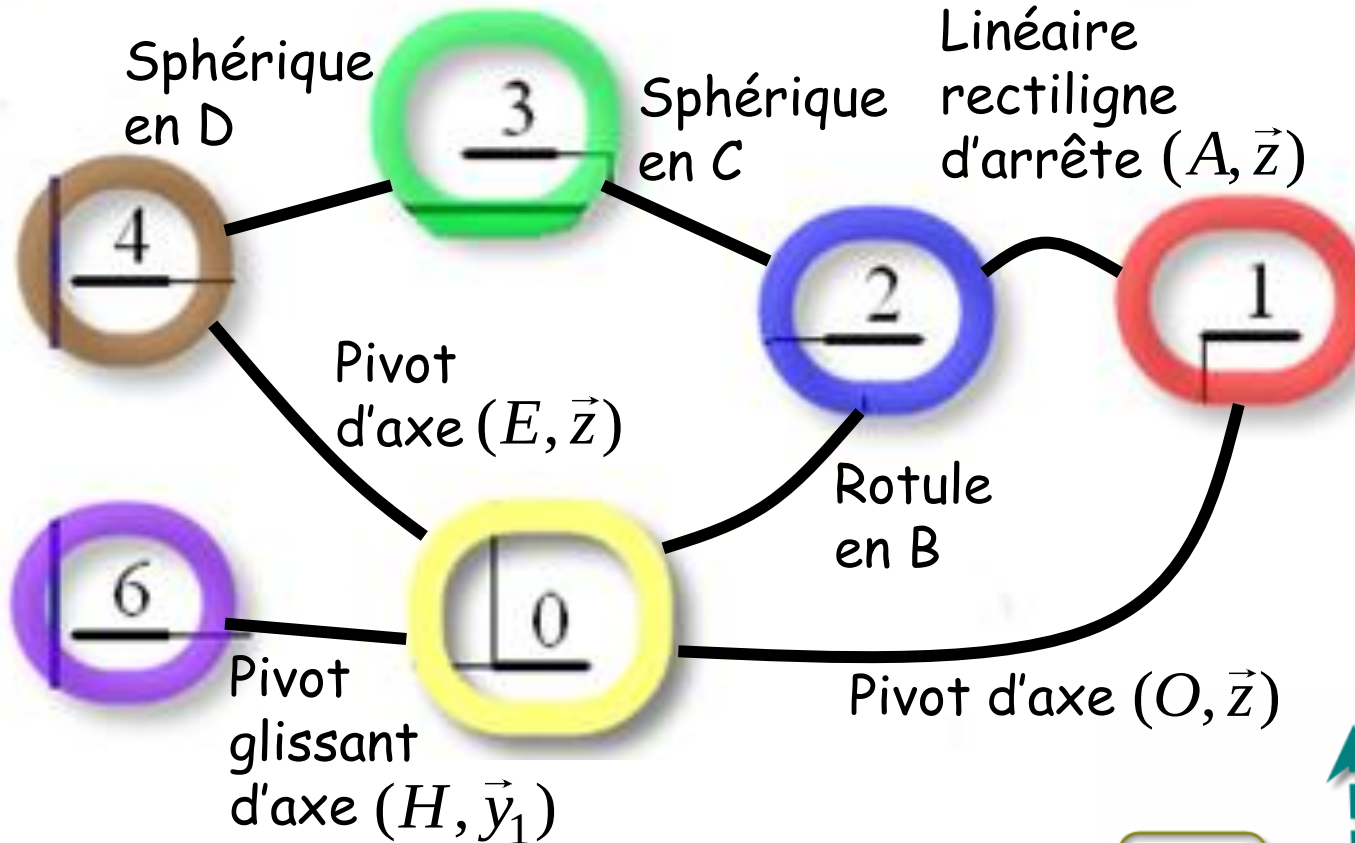
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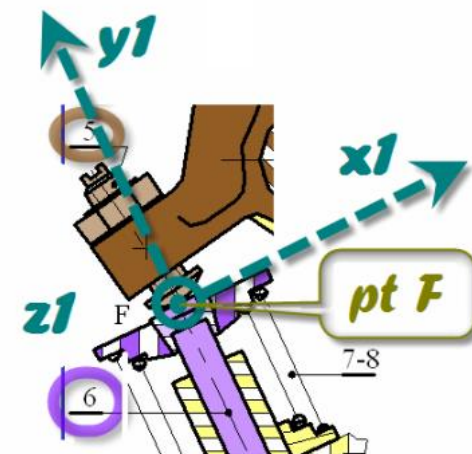
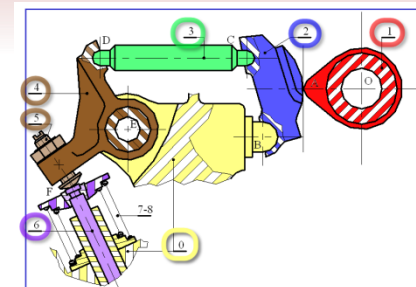
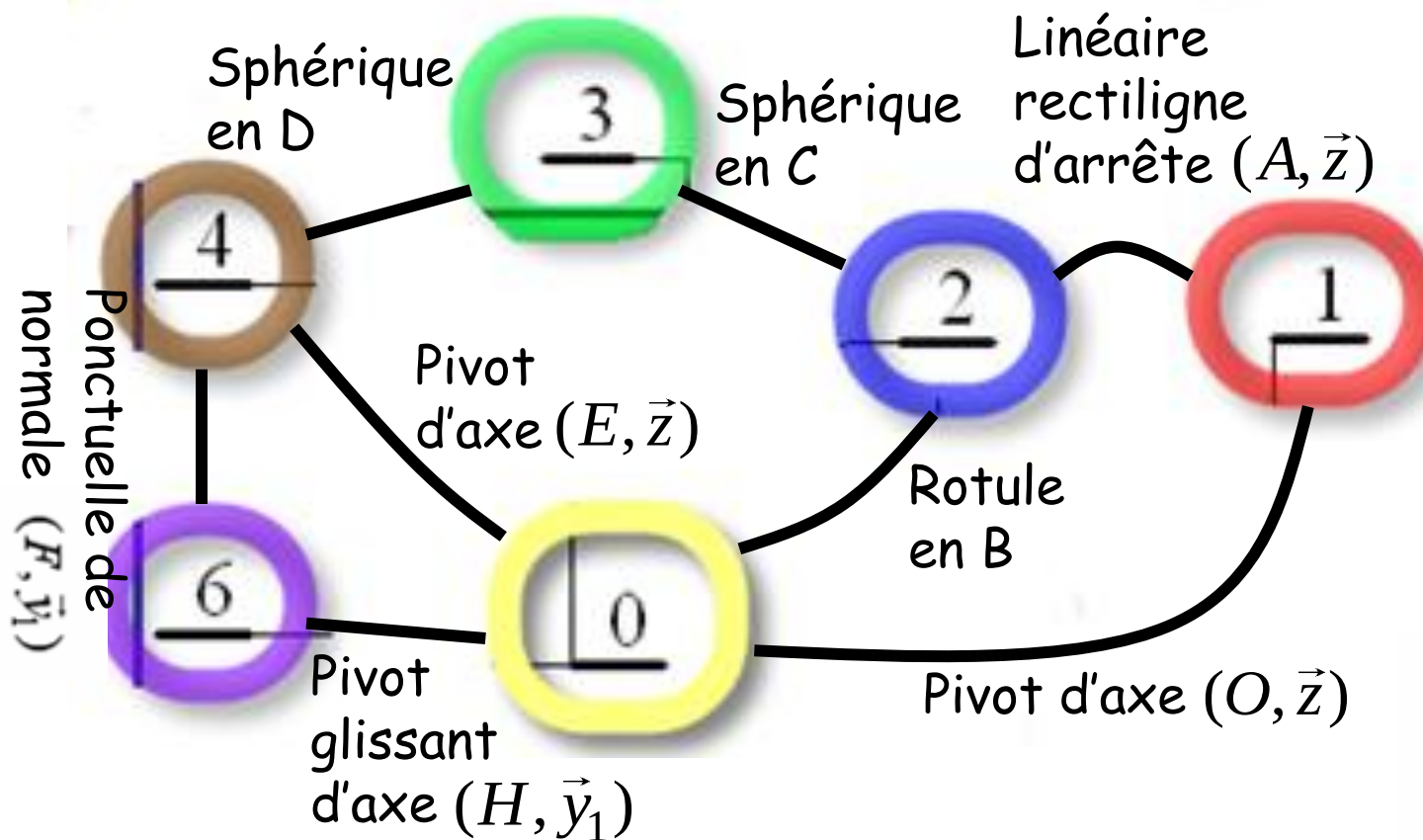
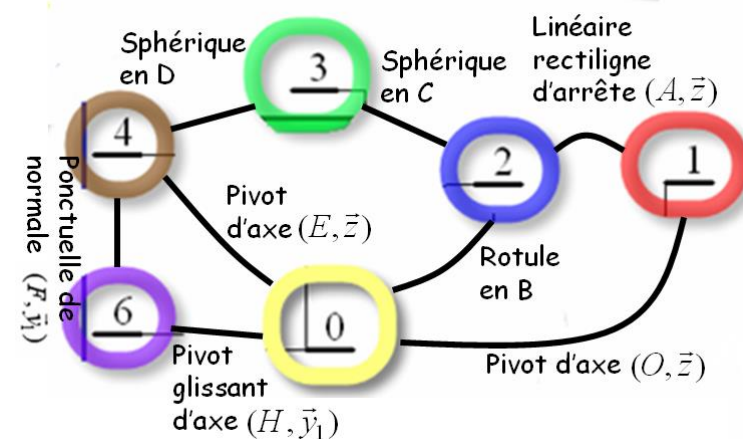
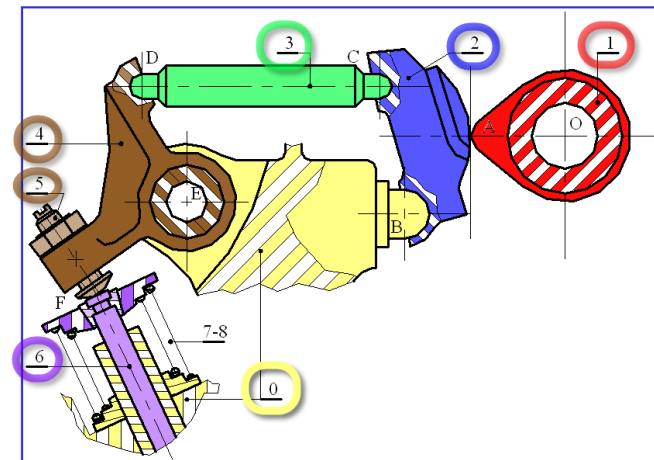
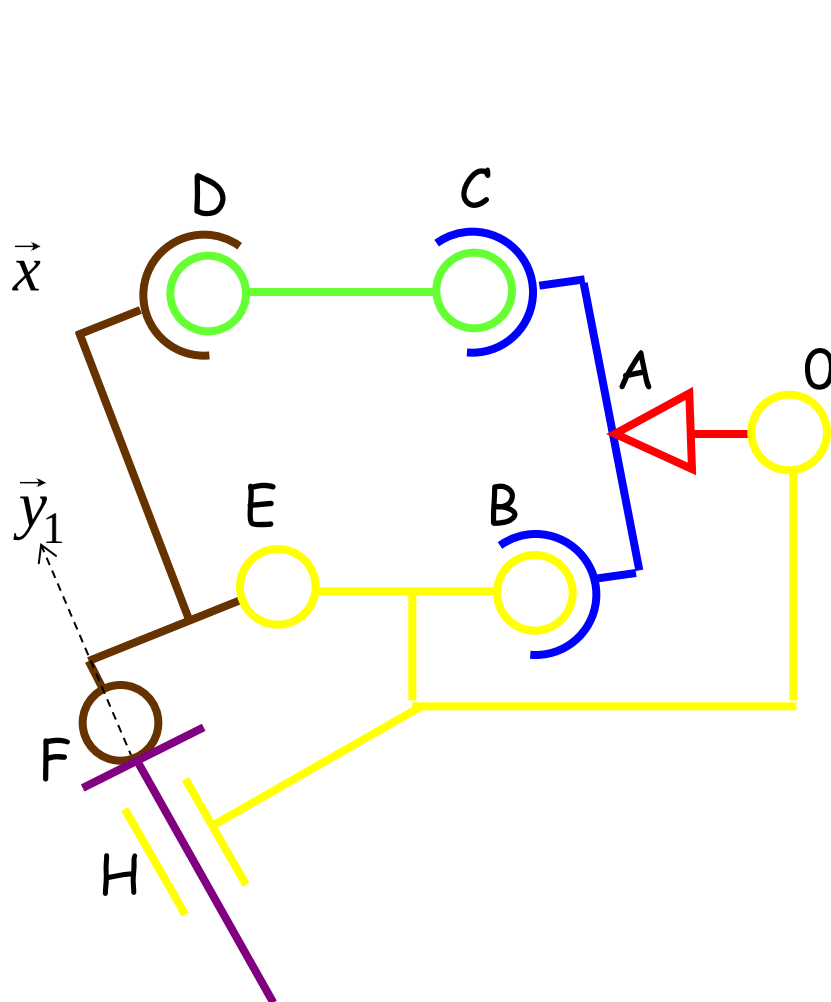
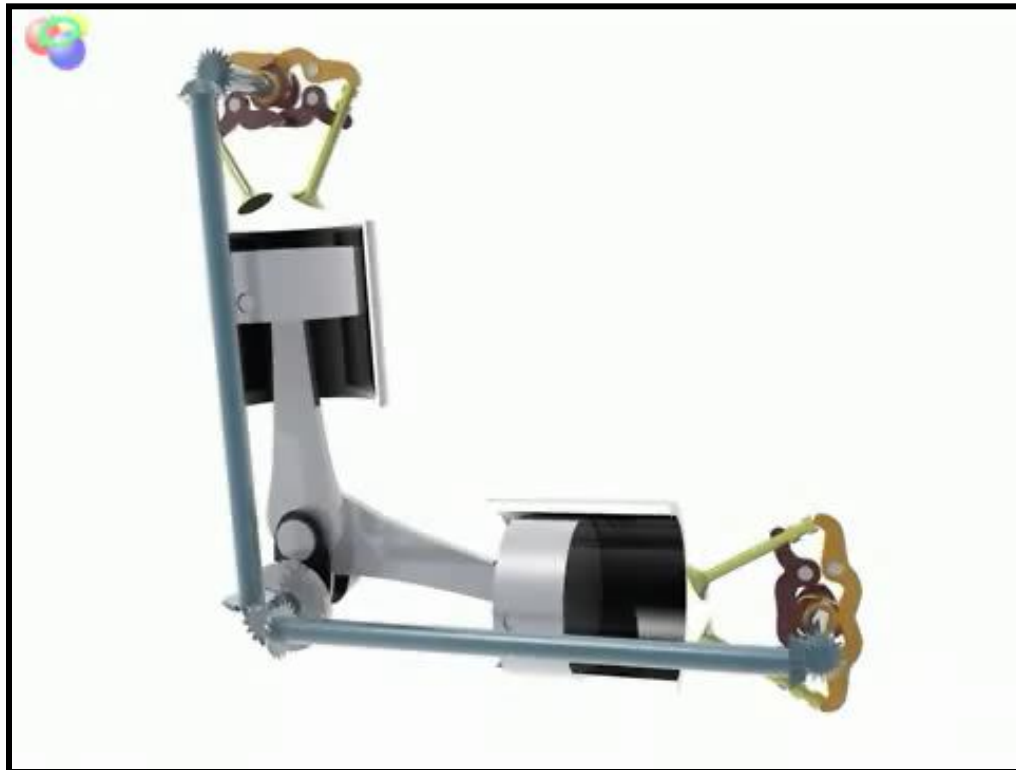


Schéma cinématique



Un autre système de commande de soupapes (Ducati) :



Ou encore le système plus classique d'arbre à came en tête...

